

To: Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), National Science Foundation

From: Rias Reed, Princeton School of Public and International Affairs, MPA Candidate '25

Subject: AI Policy Recommendations for U.S. Leadership in Competition with China

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The current framing of the 'AI race' between the United States and China misses the point. Just like the electricity race a century ago, it won't necessarily be the first nation to develop advanced technologies that will gain the advantage. Instead, the real 'winner' will be the nation who integrates this technology fastest and most effectively into industries critical to economic and strategic dominance. The U.S. is overly reliant on a defensive approach, centered on restricting China's access to advanced chips, which neglects the vulnerabilities in our own AI supply chains and underestimates the effectiveness of mid-tier chips.

This memo offers specific and actionable recommendations to shift from a reactive stance to a proactive strategy.

Recommendations:

- 1: ***Increase economic security of AI supply chains***
 - 2: ***Prioritize AI adoption within industry***
 - 3: ***Expand export controls***
 - 4: ***Protect sensitive datasets and promote secure data sharing***
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1. Increase economic security of AI supply chains

The U.S. is heavily reliant on China for the basic mechanical parts needed to run AI infrastructure. These are components like server racks, cooling units, connectors, and power supplies, which are all necessary to make AI chips function. These components are inexpensive with very low margins, which has allowed China to dominate production through decades of targeted industrial policy. If China imposes retaliatory export controls on these essential parts, it could slow down our ability to expand AI quickly and maintain our lead.

First, the U.S. should offer clear incentives to companies who bring the manufacturing of these components back home. A secondary approach would be to promote manufacturing of these components within allied partner countries in the Asia-Pacific, such as Japan, South Korea, and Australia. Targeted tax incentives would encourage U.S. and allied businesses to produce these components domestically, creating a stronger and more secure supply chain while reinvigorating the U.S.'s manufacturing sector.

Second, the U.S. should start building stockpiles of these mechanical components, which would ensure that future domestic AI infrastructure isn't vulnerable to sudden supply chain shocks. Having a reliable reserve of these parts would provide additional safeguards against potential disruptions caused by geopolitical tensions, retaliatory export restrictions, or unexpected market shortages. These stockpiles should be built gradually and sourced from multiple suppliers to

avoid drawing unwanted attention to this strategy from China.

Finally, setting regular meetings with public and private sector industry leaders within the chip and manufacturing space would allow for quicker responses to potential disruptions. Better communication and collaboration will allow us to spot risks sooner and address them before they slow us down.

2. Prioritize AI adoption within industry

Winning the AI race isn't about who develops the most advanced models first. It's about who adopts AI into their economy first and does it strategically. China understands this and is already embedding AI into manufacturing, logistics, and finance. The U.S. needs to move quickly to ensure AI adoption happens in the industries that directly impact economic power and national security.

First, AI should be deployed within manufacturing to increase efficiency and output. AI-driven predictive maintenance and automated quality control will reduce costs and strengthen production domestically. To accelerate this adoption, targeted tax incentives should be offered to manufacturers that integrate AI into their operations, allowing the U.S. to become competitive in manufacturing again.

Second, AI should be embedded into logistics operations to lower costs and contribute additional security to existing supply chains. This strategy could include AI-driven route optimization and warehouse automation. Incentivizing AI adoption in transportation and distribution operations will ensure the U.S. maintains more control over supply networks and positions itself ahead of global competitors.

Third, AI should be integrated into fintech to strengthen U.S. leadership within financial markets. AI-driven fraud detection and automated risk assessment tools will also improve efficiency and security within financial markets. Expanding regulatory sandboxes to allow fintech firms to test and deploy AI solutions faster will further position the U.S. as the leader of global financial innovation.

Fourth, AI should be used within energy and resource management. AI can significantly improve grid efficiency, predict equipment failures, and optimize energy distribution paths. These applications will reduce costs and increase reliability. Supporting AI adoption in energy infrastructure would also enhance national security by allowing power grids to become more resilient to real-time cyber threats and unexpected supply disruptions since AI's predictive capabilities allow grids to anticipate and respond to risks before they escalate.

3. Expand export controls

Current U.S. export controls focus on cutting off China's access to top-tier AI chips, but this strategy has shortcomings. Chinese companies, like DeepSeek, have shown that meaningful advancements in AI do not require creating large AI models from scratch, which is where these top-tier chips are needed most. Instead, mid-tier chips can be used to train specialized models from existing foundational models like Chat GPT or Claude. These specialized models can be

fine-tuned for specific tasks using far less computing power, allowing China to make rapid AI advancements without relying on top-tier chips.

First, the U.S. should establish a clear performance threshold for mid-tier chips and extend export controls accordingly. This would prevent chips capable of supporting large-scale specialized model training from flowing freely into China.

Second, the U.S. semiconductor firms should develop export-friendly versions of mid-tier chips, just as they've done with the restricted high-tier chips. Companies like NVIDIA have already developed modified chips with slight performance limits to comply with export restrictions while still meeting global demand. A similar approach for mid-tier chips would slow China's ability to scale specialized AI models without significantly disrupting supply chains within global markets.

Finally, these new export controls need to be aligned with U.S. allies such as Japan, South Korea, Taiwan, and the EU. Coordinated policies across major semiconductor-producing nations would close loopholes and prevent China from sourcing restricted chips elsewhere.

4. Protect sensitive datasets and promote secure data sharing

Overly broad data restrictions can slow down U.S. AI innovation and adoption, but leaving data open comes with security risks and allows competitors like China to secure an advantage. A balance must be struck between these two extremes. The U.S. needs a more calculated approach that protects truly sensitive datasets while allowing American AI firms to have the data they need to advance AI models and remain competitive.

First, the government should clearly define what qualifies as highly sensitive data and enforce strict protections for it. This may include biometric records, genomic data, and classified intelligence. These datasets could create significant national security risks if adversaries were able to access and train specialized models from them.

Second, the U.S. should encourage secure data-sharing partnerships between public and private sectors. Companies and research institutions both need large and accurate datasets to develop AI effectively, but access must be structured to prevent misuse. Establishing frameworks for sharing anonymized and non-sensitive data would allow for more domestic AI advancements.

Considerations

The adoption of AI is necessary to maintain U.S. leadership, but there are many considerations that must be accounted for in the implementation of these policies. AI integration comes with economic, ethical, and national security implications that should be weighted to ensure long-term prosperity for the nation.

1. Job displacement and workforce readiness

Expanding AI in industry will increase productivity but could also lead to significant job displacement. Policymakers should consider strategies to retrain American workers to ensure that AI-driven economic growth benefits the U.S. workforce rather than replace it. Incentives to adopt AI practices should be paired with workforce training programs to prepare workers for higher-skilled roles in AI-integrated industries.

2. Ethical implications

AI adoption has clear economic and strategic benefits, but it also raises concerns about misuse, especially within financial markets. AI-driven trading, lending, and fraud detection tools can improve efficiency, but if not properly managed, they could introduce systemic risks or be exploited in ways that significantly destabilize markets. The administration should consider targeted safeguards that reduce these risks without stifling AI development.

Rather than imposing heavy-handed regulations that slow down innovation, the focus should be on practical oversight that prevents reckless deployment. This could include ensuring AI-driven financial systems are transparent enough to detect manipulation and fraud. Significant consideration should be given to balancing the protection of U.S. financial markets while making sure AI progress isn't slowed unnecessarily.